



LOOP Range

Manual operation English

Legal notice

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Countertop Vending Machine

Model range

LOOP
LOOP XL

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1 Safety

1.1 Intended use

By using the unit as intended you will not only protect yourself, but also prevent damage occurring to the unit and its components!

The intended use of counter top coolers from the LOOP series (referred to as "unit" in the following) is to refrigerate and convey non-alcoholic beverages and their basic ingredients. CO2 or electric pumps are used as conveying equipment for the brands.

The unit is only suitable for stationary installation in a closed room. Stationary installation is to be carried out by an expert in compliance with all of the specifications given in this manual.

The inlet temperature of the liquid must not exceed 32°C otherwise the pressure in the refrigeration cycle will rise above specification.

Minimum ambient temperature 10°C, maximum ambient temperature 32°C

The intended use means that you will carry out all activities with and on the unit to the specifications provided in this document.

This unit is only to be operated by those who meet the requirements set out in this document.

Work on the unit and its components not included in the activities described in this document may only be performed by experts, means authorized and skilled people.

1.2 Improper use

Improper use of the unit and unauthorized modifications to the unit and its components may cause personal injury and equipment damage for which Cornelius Deutschland GmbH shall assume no liability. Improper use of the unit is prohibited.

The following in particular is regarded as improper use:

- Mobile operation of unit
- Operation of unit in areas without supervision by skilled personnel
- Operation of the unit in an area, where water jet is possible.
- Cleaning with a water jet
- Any handling as e.g. use or service by children under the age of 8 years
- Any cleaning or service by children without supervision by a skilled, authorized adult.
- Use by persons (including children) with physical, sensory or mental disabilities or those with insufficient experience and knowledge, unless they are supervised by a person responsible for their safety, or this person has instructed them in the use of the unit.
- Use by those under the influence of medication, alcohol, drugs or other substances which impair their physical, sensory or mental abilities.
- The refrigeration of non-specified liquids and those above the maximum supply temperatures as this may cause unacceptably high pressures to build up in the refrigeration circuit.
- Operating the unit below the minimum and above the maximum ambient temperatures (minimum and maximum temperatures *see chapter 3.3, page 15*).
- Operating the unit with conveying media other than those defined in this document.
- Operating the unit in locations and ambient conditions which do not fully meet the requirements of this manual
- Operation of the unit by untrained staff

1.3 Staff

1.3.1 Operator

The operator is the natural or legal person who uses the unit or on whose behalf the unit is used. The operator must ensure that the unit is only used as intended, in observance of the safety instructions set out in this document.

The operator must ensure that all users read and understand the safety information. The operator is responsible for the planning and proper implementation of regular safety inspections and maintenance work.

With regard to operating the unit, Cornelius Deutschland GmbH recommends observing the national regulations of the country of use which govern the operation of drink-dispensing systems.

1.3.2 User

The operator specifies who will operate this unit. Cornelius recommends the following:

- If this unit is only to be operated by employees, they are to be instructed in its use, demonstrate their abilities to use it to the operator or their authorized representative, and be expressly charged with its use. This document is to be available to staff at all times.
- If this unit is openly accessible and set up so that untrained staff can use it, the operator is to provide instructions for use directly at the unit; these must be clearly understood by this group of people, therefore ensuring that the unit will be handled safely.

1.3.3 Expert

An expert in terms of this document refers to someone who has the relevant training, experience and information and knowledge of relevant standards, laws, regulations, accident prevention regulations, generally accepted safety-related regulations and operating conditions to be able to perform the required activities as well as recognize potential risks and avert them.

For assignments requiring expert knowledge, e.g. in electrical engineering, mechanics or fluid technology, only skilled workers with the right qualifications are to carry these out.

An expert must also have received technical training in the unit-specific special features of Cornelius products. The assigned tasks are always to be carried out in compliance with this manual.

1.4 Presentation of warning

The documents supplied with the unit provide warnings regarding any hazards that might exist.

1.5 Safety instructions

1.5.1 General safety information

Any work on the unit and its components which goes beyond the operation and servicing and maintenance that the operator is authorised for, may only be performed by experts (for a definition of experts, see chapter 1.3.3, page 5). Furthermore, it is crucial that when performing work on the unit all safety information is observed; this information is set out in the following sections. Some of the tasks may have additional safety information which highlights the specific hazards associated with such work.

1.5.2 Disclaimer of liability and warranties

If work is undertaken on the unit which is not described in this document, Cornelius Deutschland GmbH shall not assume any liability for any resulting hazards and damages. The same applies to described work which is partly or not fully performed in compliance with the regulations set out in this document.

Warning!**Risk of personal injury and equipment damage due to improperly executed work**

Improperly executed work at the unit will cause dangers to persons and damage to the unit.

- Have all work at the unit carried out by Cornelius Deutschland GmbH or by a service partner.

Warning!**Risk of personal injury and equipment damage due to the use of non- approved spare parts and accessories**

Using spare parts and accessories that are not recommended by the manufacturer may lead to personal injury and equipment damage.

- For your own safety and to protect your warranty, only use original spare parts.

1.5.3 Safety information to prevent personal injury and equipment damage

Please always observe the following safety information in order to prevent personal injury occurring:

Danger!**Risk of death from electric shock**

Touching live electrical parts will result in a risk of electric shock!

- Do not carry out any work on the electrical system.

Danger!

The unit is not intended for any immersion in water for any reasons (e.g. cleaning)

Risk of electric shock!

Warning!

When positioning the appliance, ensure the supply chord is not trapped or damaged.

Warning!

Do not locate multiple portable socket-outlets or portable power supplies at the rear of the appliance.

Warning!**Risk of burns when touching hot parts of the unit**

Touching parts of the unit after it has been in continuous use over an extended period of time will result in a risk of burns.

- Take appropriate safeguard measures, such as by wearing heat- resistant protective gloves.

Danger!

Risk of poisoning and risk of explosion due to improper handling of CO₂ cylinders

Risk of death from CO₂!

- Observe all information on occupational safety for the safe operation of dispensing systems as applicable in the respective country of installation.

Danger!

Risk of personal injury and equipment damage due to improper operation

Risk of death from improper operation!

- Make sure that only users who are users as defined in the user section use the unit; see *chapter 1.3.2, page 5*.
- Children must be supervised to ensure that they do not play at or with the unit.

Danger!**Risk of personal injury and equipment damage due to leaking refrigerant**

In accordance with DIN EN 378 standard, part 1, the refrigerant used with this unit, R290 (propane), is classified as safety group A3. It is not poisonous and not caustic or corrosive. However, it is combustible and highly flammable. As a result, a potentially explosive atmosphere may occur in the case of leaks.

The refrigerant is heavier than air and, consequently, accumulates at the bottom and may enter lower lying areas (e.g. sewage systems). At high concentrations, this may lead to a lack of oxygen, resulting in a risk of suffocation. Lower concentrations may have a narcotic effect. The refrigerant has a very slight, somewhat sweet smell.

- Avoid any damage to components that carry the refrigerant
- If you suspect any leaks, however small, shut down the unit, air the room well and inform your service partner.
- Do not allow direct contact with the skin of any leaking refrigerant.
- Fire, open flames and smoking are prohibited.
- If you begin to notice any smell, leave the room and alert the fire service

Warning!

Do not store explosive substances such as aerosol with a flammable propellant in this appliance.

Attention!

Do not store explosive substances such as aerosol with a flammable propellant in this appliance.

Notice!

In particular, safety standards are to be observed in their scope of validity (e.g. IEC 60335-2-75, EN 60335-2-75).

Draw also attention to any regional rules that may be applicable.

2 Transport and packaging

Choose a suitable packaging when returning the unit itself or one of its components to Cornelius Deutschland GmbH, e.g. for repairs. In particular, make sure that the unit and any components are protected from shock/impact, moisture, dirt and electrostatic discharge (ESD). This will prevent transport damage to the unit and to the components, for which the manufacturer shall assume no liability.

Caution!**Component damage due to freezing liquids**

Ambient temperatures that are below freezing will lead to the freezing of any water or cleaning agent residue remaining inside the unit. This will lead to damages of internal components.

- Before shipment, storage or relocation of the unit, the unit is to be cleaned and the cleaning solution is to be fully drained from the unit.

2.1 Storage

Avoid excessive temperature fluctuations as condensate may form, which in turn may cause damage to the unit or to the components.

The permissible storage temperature is -10 °C to +50 °C.

The acclimatization period is 6 hours.

Caution!**Damage due to improper storage**

Dirt or moisture entering a unit, as well as certain weather conditions (e.g. condensate forming at the unit, sunlight) will cause damage to the unit and its components.

- Protect the unit and its components by storing the unit in a clean and dry place, and by ensuring stable ambient conditions.
- If possible, store the unit in its original packaging. Unpacked units must be covered with a dustproof cover. No condensate must form under the cover.

Caution!**Risk of electrostatic charge!**

Improper handling or storage may result in electrostatic charges.

- If possible, store units and/or any electronic components in their original packaging.
- Keep units and/or electronic components away from charged objects, fields and insulators.
- Avoid electrostatic charges when removing packaging and/or handling electronic assemblies and components by working at an ESD protected workstation or work area.
- When working at the unit or its components wear a grounding (antistatic) wrist strap at the very least and wear antistatic gloves if necessary.

Caution!**Component damage due to material ageing**

Material can age due to long storage periods, thereby affecting the material's properties (e.g. plastics and seals may become brittle). The properties of lubricants may change due to long storage periods.

- Check the assemblies and components for damage before each use/before installing them. Do not install assemblies or components that show visible signs of ageing.

Caution!**Component damage due to freezing liquids**

Ambient temperatures that are below freezing will lead to the freezing of any water or cleaning agent residue remaining inside the unit. This will lead to damage to internal components.

Before shipment, storage or relocation of the unit, the unit is to be cleaned and the cleaning solution is to be fully drained from the unit.

3 Disposal

Disposal of the units must be carried out in compliance with the applicable local and/or national and international regulations. Units must not be disposed of with household waste.

If the unit contains fuels or lubricants in liquid, paste-like or gaseous form, such as oil, grease, cooling agents etc., such fuels or lubricants are to be collected using appropriate measures and disposed of in compliance with the applicable local and/or national and international regulations. Such fuels or lubricants must always be prevented from seeping into the ground, the sewage system and any bodies of water, and must always be prevented from entering the atmosphere.

4 Installation Requirements

4.1 Installation Sites

The unit is intended to be installed in i.e. staff kitchen, shops, offices, hotels, motels, other residential environments, bed and breakfast and cantinas.

Comply with the valid national regulations for installation sites and electrical connections. Ventilation of the installation sites must be appropriate for unit output. Inadequate ventilation of the unit will result in overheating and damage. Always make certain that no intake or discharge vents are covered or blocked.

	LOOP	LOOP XL
Heat output in watt	660	1000
Air flow in m ³ /hour	150	300

4.2 Electrical Connections

A socket outlet with an earthed contact featuring a maximum protection of 16 amperes is required.

Caution!

The unit is not to be installed with a fixed wiring to the electric mains. Only the plugged connection to a socket outlet is allowed.

The line voltage must always be within the following tolerances: 230 VAC +6%/-10% / 50 Hz

	LOOP		LOOP XL	
	PEM	POM	PEM	POM
Power consumption in watt	315	365	500	600

Should placement of the unit require an extension cable, please note the following points:

- Cable length must not exceed 50 meters.
- Only use cable with a size of 2.5 mm².
- Only one unit is allowed per extension cable.

4.3 Hydraulic system components

Attention!

For prevention of backsiphonage of water from the unit into the water mains a backflow preventer is mandatory to be installed directly on the water inlet connector of the unit.

Attention!

In case of disconnection of the flexible water hose set in the water mains due to any purposes, it is mandatory to replace it be a new hose-set for reconnection of the unit to the mains water.

4.4 System accessories

This unit is intended to be installed as a system with following accessories by Cornelius or equivalents.

Mains water:	Water filter	PN 13-7796-400
CO ₂ :	Pressure reducer	PN 14-9535-202
Syrup:	Pump plate	PN 14-8712-000

5 Installation

The unit must be installed by a trained service technician.

Please take care, that the socket for the unit is always accessible.

If the power supply cable to the unit is damaged, it has to be replaced by the manufacturer, the service partner or any other qualified person to avoid safety hazard.

The unit must be planar aligned. Max. angle of $\pm 2^\circ$ is permitted.

5.1 Premix

5.1.1 CO₂ Connection for Premix

You will require a one-stage pressure regulator, 0,7 MPa (7 bar). Use tubing with 4 mm ID (inner diameter) to connect the pressure regulator with the CO₂-distributor and the beverage containers.

5.1.2 Connecting Premix Container

Use minimum 6 mm ID tube to connect beverage container with correct cooling coil into the unit.

5.2 Post-mix

5.2.1 Water Connection for Post-mix

The connections to mains water are not delivered with this unit. Please use only water connections that are in line with IEC/EN 61770. Connect the unit to a feed line with an inner diameter of 8 mm. We recommend use of a water filter and a water pressure regulator for the water input. To permit flushing of the filter, a t-piece should be mounted downstream of the filter. The water flow pressure should be 0,2 MPa (2 bar) (mount control pressure gauge on water pressure regulator). The unit must only be operated with a water pressure regulator.

5.2.2 CO₂ Connection for Post-mix

You will require a two-stage pressure regulator, 0,7 MPa (7 bar).

Use tubing with 4mm ID.

One pressure for the carbonator bowl and the other one for the syrup flow.

Set the CO₂-pressure: see table on page 21 (chapter 6.1).

5.2.3 Connecting Post-mix Syrup

Use minimum 6 mm ID syrup tube to connect syrup container with correct cooling coil into the unit.

6 Putting into and out of Operation

6.1 Putting into Operation

Before starting operation of the unit the transport lock must be removed.

In order to ensure the proper function of the unit, please follow the below outlined points in order.

Comply with the cleaning regulations defined by local laws before beginning each operation.

Connectors must be cleaned every time prior to attaching a beverage container.

Connect coupling to beverage container.

For soft drink note: gray coupling = CO₂, black coupling = beverage/syrup

Open the valve of the CO₂-cylinder and the pressure regulator. Check the CO₂-pressure on the pressure regulator.

It should be within the following standard values:

syrup:	0,15-0,25 MPa (1,5-2,5 bar)	cold carbonation
carbonator pressure:	0,35-0,40 MPa (3,5-4,0 bar)	
diet/light-beverage:	0,10-0,15 MPa (1,0-1,5 bar)	

Set the CO₂-pressure by turning the control screw:

- To increase the pressure: Clockwise
- To reduce the pressure: Counterclockwise

Afterwards check the CO₂-lines for leaks by closing the CO₂-valve. The pressure (high pressure gauge) displayed at the pressure regulator should not drop. If it does, notify the service technician immediately! Do not forget to re-open the CO₂-valve if no leakage is evident.

Open the water feed line and check the flow pressure. Standard value: 0,2-0,25 MPa (2,0-2,5 bar).

Set flow pressure at the control screw on the water pressure regulator:

- To increase the pressure: Clockwise
- To reduce the pressure: Counterclockwise

Check the beverage/syrup lines for leaks. Only a visual inspection is possible. If liquid is leaking, call a service technician.

Close the water feed line. The pressure display should not drop. If it does, notify the service technician immediately! Check the incoming waterline for leaks. Do not forget to re-open the water feed line if no leakage is evident.

6.2 Turning on the Unit

The water bath must be filled to the overflow with tap water. Refer to the technical data for the amount required.

Take care, that no water runs into the agitator motor. Reference: Use for this a suitable funnel and filler tube.

To prevent algae from forming in the water, add the disinfectant Molco (PN 14-9670-150). The 150 ml container of disinfectant is sufficient for 30 liters of water.

Insert the mains plug for the cooler into the socket outlet with an earthed contact.

Units with water bath are ice bank controlled. These units start working automatically after the water bath is filled with water. The compressor will cycle off after the ice bank has been built. (full ice bank capacity). The control board of the unit has a time delay for switching on and off the cooling system when it runs in

ice bank mode. After the cooling system is switched on the running time is no less than 5 minutes. Switch off signals will be ignored in this time. After the cooling system switches off, the delay is no less than 3 minutes. Switch on signals will be ignored in this time. The 3 minutes delay occurs when turning on the unit and after a break in power supply. This unit contains a 2-pin ice bank probe.

Post-mix units include a carbonator pump which switches on automatically and fills the carbonator bowl. The carbonator pump switches off when the water has reached its highest level in the carbonator bowl but after no more than 5 minutes. Long run periods are signs of leaks or insufficient water. It is then only possible to turn the pump back on by executing a network reset (pulling out the mains plug briefly).

The carbonators in post-mix units are protected against running dry. If the flow rate quantity is so large that the "empty electrode" is activated, then the carbonator bowl is empty. In such cases voltage supply for the post-mix valves will be switched off. A beverage withdrawal is only again possible if the carbonator bowl is filled.

Release air from the carbonator bowl by pulling the safety valve for about 2 to 4 seconds.

6.3 Purging the Syrup Systems

1. Place a cup under the nozzle.
2. Manually open each syrup valve in turn and observe syrup flow. Purging is completed when all air is removed from the system. If syrup continues to show air bubbles, check tightness of line connections.

6.4 Adjusting the Water-to-Syrup Ratio for Post-mix

Adjust the water flow rate on the post-mix valve in accordance with the valve supplier specifications.

Note:

Allow the refrigeration unit to operate until minimal amount of ice has formed on the evaporation coils, making this adjustment. Alternatively, allow the full ice bank to form (compressor cycle off) before adjusting the ratio. The drink temperature should not be higher than 5°C.

The water-to-syrup ratio is measured by use of a ratio cup having two chambers into which the soda and a selected syrup are dispensed. The ratio, if incorrect, is adjusted by turning the related syrup flow regulator adjusting screw located near the solenoid valve.

- To increase the syrup flow rate: Clockwise
- To reduce the syrup flow rate: Counterclockwise

6.5 End of Operation (End of dispense-time)

It is imperative that the CO₂-cylinder and water line be turned off each time operation is ended. Be sure to open again before continuing at operation.

6.6 Daily Inspection

- Check whether carbon dioxide and water lines are open.
- Check the beverage/syrup lines for leaks. Only a visual inspection is possible. If liquid escapes, call a service technician.
- Check the CO₂-lines for leaks by closing valve on the CO₂-cylinder. The inlet pressure indicated on the pressure regulator should not drop. If it does, call a service technician immediately.
- Do not forget to re-open the CO₂-cylinder valve afterwards.

6.7 Taking out of operation (Vacation, end of season)

Perform the following steps in case of protracted standstill periods:

- Close the CO₂-cylinder, the CO₂-stopcocks on pressure regulators and the water feed line.
- Pull the mains plug out of socket outlet with earthed contact.
- Detach the couplings from beverage containers.
- Have the system emptied and cleaned.

Only trained specialists are to carry out this procedure.

7 Cleaning and Disinfection Directions

7.1 Cleaning Directions

Comply with the valid national regulations for cleaning bar equipment at the particular installation site.

Clean connection parts and tap fittings in advance whenever making connections or changing the type of beverage. Clean parts coming into contact with air and beverage (e.g. dispense nozzle), on a daily basis.

The condenser fins must be cleaned in regular intervals. These vary according to the amount of dirt in the air at the installation site (approximately every three months). This is best done with a brush and a vacuum cleaner.

The water level in the water bath must be checked regularly and the contents must be exchanged at least once annually. Algae formation can be reduced by adding disinfectant (order number 14-9670-150). For a 30 liters water-bath one disinfectant container is sufficient.

The unit is to be emptied and cleaned by trained person only based on the following recommendations:

To be cleaned by trained specialists	CO ₂ -lines	Product-lines	Syrup-lines	Sodawater-lines
Before commissioning		X	X	X
Before each change of type of beverage		X	X	
Before and after a pause of more than 1 week		X	X	
Every 2 weeks		X		
Every 3 months			X	X
Every 12 months	X			

7.2 Cleaning and Disinfection Procedure before use

In order to achieve a proper hygienic performance of the dispense equipment, it is crucial to run the initial and recurring sanitization procedure (intervals according to DIN 6650-6) on all product and water lines of the system.



Attention !

Cleaning / sanitizing agents are harmful and may cause severe health injuries !
During the work with any agents make sure to always wear proper clothing
Take care of an adequate behaviour towards hygiene while working on the equipment (e.g. disinfecting hands prior to work, etc) in order to professionally deal with the matter.
The unit should be cleaned / sanitized starting as close as possible from the mains water connection (wall outlet), to make sure that also the tubing is being treated.

Remark -> Water filters :

In case the system is equipped with a water filter, the filter cartridge needs to be removed before doing the cleaning / sanitization and replaced by a blind plug that allows bypassing the filter. Do not use any empty service filter cartridges to flush the sanitizing agent into the system, as this will not allow a proper and consistent level of sanitizing agent in the unit.

In addition, the high levels of agent passing through the dispenser when using such cartridges may damage components such as sealings, O-rings, etc. in the dispenser.

Remark -> Carbonator- / Circulation pumps

With units such as Triton, Apexx and Energize the carbonator- and soda circulation pumps must be turned off during the cleaning / sanitization process (otherwise foaming issues will occur).

Remark -> Stillwater lines inside the cooler

In case your equipment has still water lines, it must be made sure that these are also being cleaned / sanitized.

In case your equipment currently does not use Stillwater, the line must be equipped with a stopcock to manually drain sanitizing agent from this tube.

In case the Stillwater line is in general not being used on the cooler, it is recommended to disconnect this line as close as possible from the water cycle and close the water line with a blind plug (this avoids areas within the circuit which might not be sufficiently flushed).

Remark -> Electrical post-mix valves, which can not be manually operated

These type of taps must be operated electrically. When doing this, it needs to be considered that on some dispense equipment the 24V electrical power supply to the valves is cut, when the empty electrode in the carbonator bowl is reached. In that case the empty sensing must be bypassed, by e.g. short circuiting all connectors on the plug of the level probe connection.

Remark -> CO₂- or Water pressure sensors on the dispense equipment

Depending on the type of equipment you may have CO₂- or water pressure sensors installed, which will in case of low CO₂ or low water supply cut the 24V power supply to the valves.

In order to still be able to operate the system during the sanitization process, it is required to short circuit such sensors (make sure to put them back into operation after service).

Remark -> Post-mix valve blocks

It is recommended to clean / sanitize the valve blocks separately. Especially valve blocks with an integrated stop cock (e.g. Lancer block) need special care, as the gap in the stop cock allows bacteria to collect, which might not be sufficiently treated with the sanitizing agent.

In case the dispenser is out of operation (without cooling) for more than 10 days, it is necessary to run the cleaning- and sanitization procedure prior to putting the unit back into operation.

Should it be necessary, to open a once sanitized system again (e.g. to install a new water filter) all opened connections must be disinfected with a sanitization spray prior to reconnecting.

A.) Flushing-in of sanitization agent (e.g. P3 Ansep CIP from Ecolab)

1.) Syrup Lines

- 1.1) The product lines filled with syrup must first be flushed with water. For this an external water distributor can be used to connect and flush several syrup lines simultaneously. The post-mix valves should be operated on the syrup side only for flushing.

Attention ! Gas driven syrup pumps must not see any positive pressures on the incoming side of the pump, as this may damage the pumps.

- 1.2) Connect the water distributor to the CO₂ supply in order to drive the water out of the syrup lines. This avoids that the sanitizing agent is being diluted in the tubings. Afterwards close the CO₂ gas and depressurize the complete system again.
- 1.3) Fill your cleaning tank with the cleaning / sanitizing agent according to the mixing ratios given by the manufacturer of the agent.
(e.g. when using P3 from ECOLAB -> 50ml for 1 litre water = 5% solution)
In order to achieve a proper mixing in the cleaning tank, it is recommended to first fill in the agent and then top-of with clean water.
- 1.4) Connect the syrup lines to the cleaning tank and dispense cleaning / sanitization agent from every single dispense valve. Make sure that agent leaves from all installed taps.

2.) Water Lines

- 2.1) Close the water- + the CO₂ supply to the unit and depressurize the system with the taps
- 2.2) Disconnect the water supply to the unit and cut the power supply to the carbonator pump in order to avoid dry running of the pump (e.g. by pulling the plug of the pump).
- 2.3) Connect the water line feeding the dispenser to CO₂ gas pressure and drain all water from the tubing of the system. This ensures that the sanitizing agent that is afterwards flushed into the dispenser is not being diluted e.g. in the carbonator bowl. Afterwards close the gas supply again and depressurize the system.
- 2.4) Connect the cleaning tank to the water inlet of the dispenser and pressurize the tank with approx. 3 bars pressure (Attention ! Remove any water filters before doing this !).
- 2.5) Pull the safety relieve valve on the carbonator bowl and carefully let some sanitizing agent leave the valve. This ensures, that the carbonator bowl is flooded completely with sanitizing agent all the way to the top.



Attention ! : Avoid that excessive amounts (= >50ml) of agent is being introduced into the water bath, as this will carry the risk of corrosion of metal parts in the water bath. In case larger amounts were spilled, the water in the water bath needs to be replaced.

The LOOP post-mix cooler works with a membrane / diaphragm pump please make sure the pump is in operation when the sanitizing agent is being flushed into the unit. This will ensure that all parts of the pump will come into contact with the sanitizing agent.

- 2.6) Flood the complete dispense system with the cleaning / sanitizing agent by operating the post-mix valves. Ensure, that on all valves clearly visible agent is being drawn.



Effect- / working time for the cleaning / sanitizing agent is min. 20 minutes !!

In the meantime remove the dispense nozzles from the valves and sanitize them manually by putting them for 20 minutes into sanitizing agent.
In case it is seen that the nozzles are heavily dirty, clean the nozzles mechanically by using a clean brush and sanitization agent.

B.) Flushing-out of sanitization agent

1.) Syrup Lines

1.1) Disconnect the cleaning tank and bring out any remaining sanitizing agent from the syrup lines by using CO₂ gas. Afterwards close the CO₂ supply and depressurize the system via the dispense valves.

1.2) Connect the water distributor and sufficiently flush clean water through the syrup lines (for this please open only the syrup side of the dispense valves).

In case a hygiene water filter is being used with the dispenser, it is recommended to use for this flushing the water coming from the hygiene filter.



Attention ! It must be ensured, that no cleaning / sanitizing agent remains in the dispense system after service (risk of health injuries) !
The prove that all agent residues have been removed must be tested with indicator or test papers (contact agent manufacturer) & must be documented

1.3) Connect the syrup containers again to the syrup lines and dispense syrup on the post-mix valves until a consistent flow of syrup occurs again.

2.) Water lines

2.1) Pressurize the water line to the cooler with CO₂ pressure and dispense all sanitizing agent from the valves. This makes it easier to flush out any remaining agent from the system.

2.2) Close the gas again and depressurize the unit via the dispense valves.

2.3) Install a new water filter cartridge and reconnect the unit via the filter to the mains water line again (refer to filter manufacturer guidelines in case the filter needs priming prior to use).

In case a hygiene filter is being used on the dispenser, it must be ensured that the new filter cartridge is inserted prior to flushing the system with water. This ensures that a just sanitized system is not being contaminated again by using poor quality mains water to flush out any remaining sanitizing agent.

Spray the filter head and the connecting position of the filter cartridge with an adequate sanitizing spray to avoid any introduction of bacteria again.

2.4) Open the mains water supply. Pull the safety relieve valve on the carbonator bowl until only clean water leaves the valve, to ensure that there are no residues of sanitizing agent left in the head area of the carbonator bowl.

2.5) Open the CO₂ gas supply to the carbonator and reconnect the carbonator pump to the power supply.

2.6) Dispense sufficiently water from the post-mix valves to ensure that no sanitizing agent is left in the system.

Depending on the unit type (Over counter dispenser / large soda circuit installation with long python runs, etc.) the amount of water that needs to be dispensed may vary.



Attention ! It must be ensured, that no cleaning / sanitizing agent remains in the dispense system after service (risk of health injuries) !
The prove that all agent residues have been removed must be tested with indicator or test papers (contact agent manufacturer) & must be documented

Dispense from each tap a few beverages to fully put the system back into operation again. The cleaning / sanitization procedure must be documented accordingly and the documentation must remain at the cooler (this may vary depending on local regulations).

8 Problems and Troubleshooting

Before seriously troubleshooting the dispense equipment, first check:

- Is the electricity to the unit interrupted?
- Are the beverage containers empty?
- Is the CO₂-cylinder empty?
- Are all valves on CO₂-cylinder open?

Type of problem	Cause	Remedy
Beverage too warm while compressor is running	Condenser dirty or covered	Use brush to clean condenser between louvres
	Too much beverage being removed	Examine drink-draw capacity
Beverage too warm, compressor not running	Not enough water in water-bath	Fill up water level
Beverage foams at a tap	Electronic control or compressor defective	Call service technician
	Product/syrup stored too long and enriched with CO ₂	Connect container with fresh basic material
Beverage foams at all taps	CO ₂ -pressure too high	Set pressure
	All products/syrup enriched with CO ₂	Connect container with fresh basic materials
	All beverage too warm	Check storage temperature see "Beverage too warm"
Valve only pours syrup (only post-mix)	Fault in water circulation	Check if water feed line is open
		check water flow pressure of 0,2 MPa (2 bar)
		Check if the carbonator motor is running, if not call service technician
	Valve defective	Exchange valve
CO ₂ -volume in the beverage to low (flat drinks)	Water lines frozen	Call service technician
	Air in carbonator (only post-mix)	Bleed air
	CO ₂ -pressure too low	Adjust pressure
Too much or not enough syrup in beverage (only post-mix)	Regulator in tap is clamping	Call service technician
	Delivery pressure for syrup too low or too high	Adjust CO ₂ -pressure

9 Technical data

	LOOP		LOOP XL	
	PEM	POM	PEM	POM
Output capacity soft drink 0,3 liters				
Peak dispense at 2 drinks per minute	100		180	210
Continuously – drinks per hour	50		80	75
Supply voltage	230V/50Hz		230V/50Hz	
	PEM	POM	PEM	POM
Power consumption in Watt	315	365	500	600
Compressor output in Watt (HP)*	250 (1/5)		395 (1/3)	
	PEM	POM	PEM	POM
Cooling performance in Watt	350		600	540
Weight of ice bank in kg	5.3		7	
Ice bank capacity in kcal	424		560	
Ice build up in minutes	110		110	
	PEM	POM	PEM	POM
Water bath capacity in litre	14.5	13	20	19
Dimensions in mm				
Height (soft drink)	505		550	
Width	355		415	
Depth	625		660	
	PEM	POM	PEM	POM
Shipping weight in kg	37	38	46	50

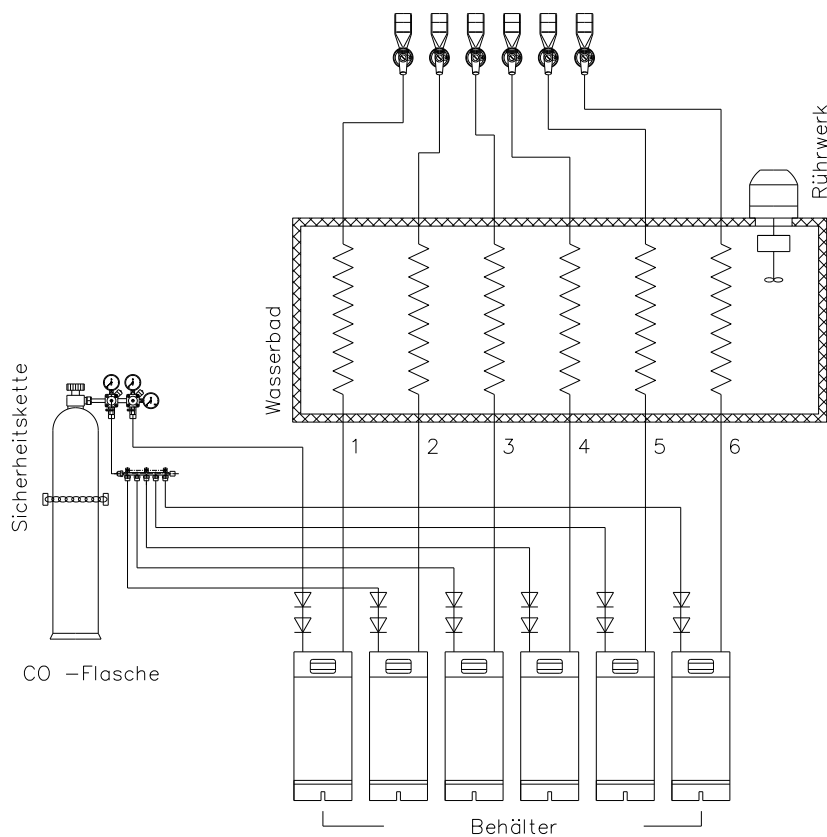
* at -10°C evaporation temperature

Cooling capacities and output capacity for soft drinks at 24°C ambient temperature and water or syrup inlet temperature of 24°C and beverage outlet temperature of less than 5°C.

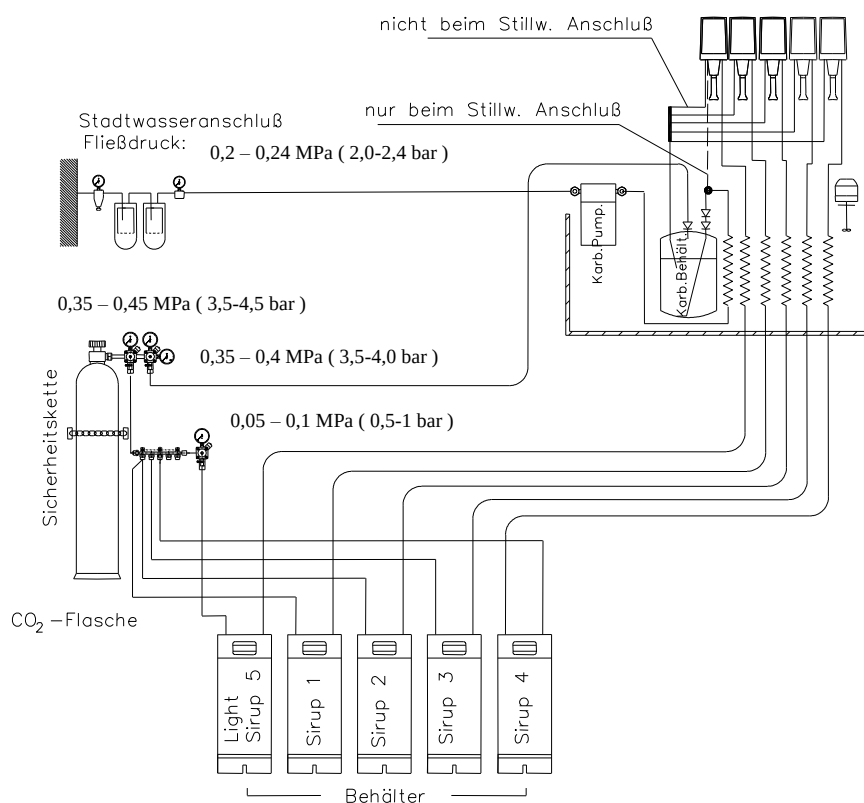
We reserve the right to make modifications.

10 Flow scheme

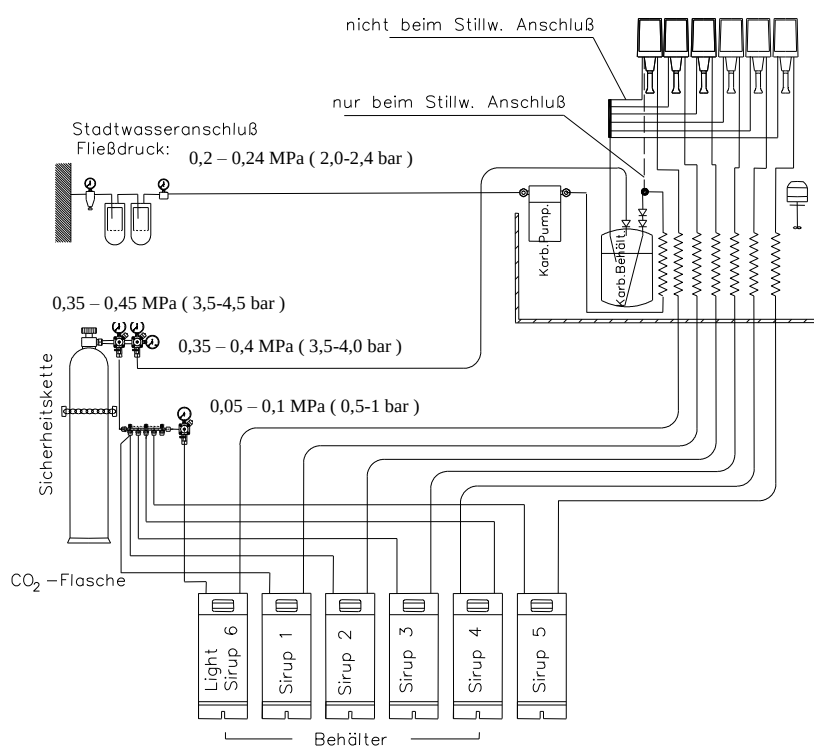
LOOP/LOOP XL 6 Valve Premix



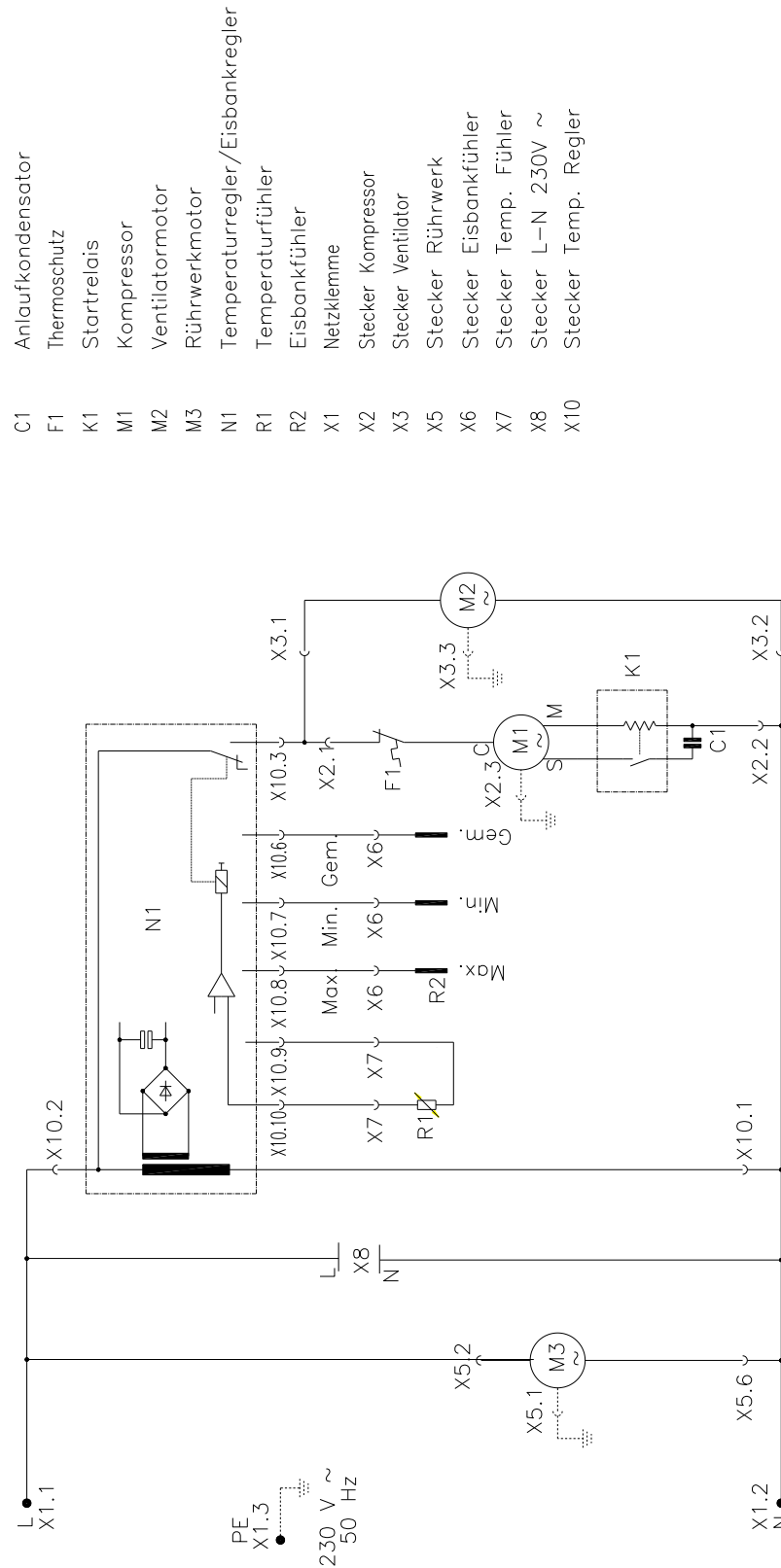
LOOP 5 valve Postmix



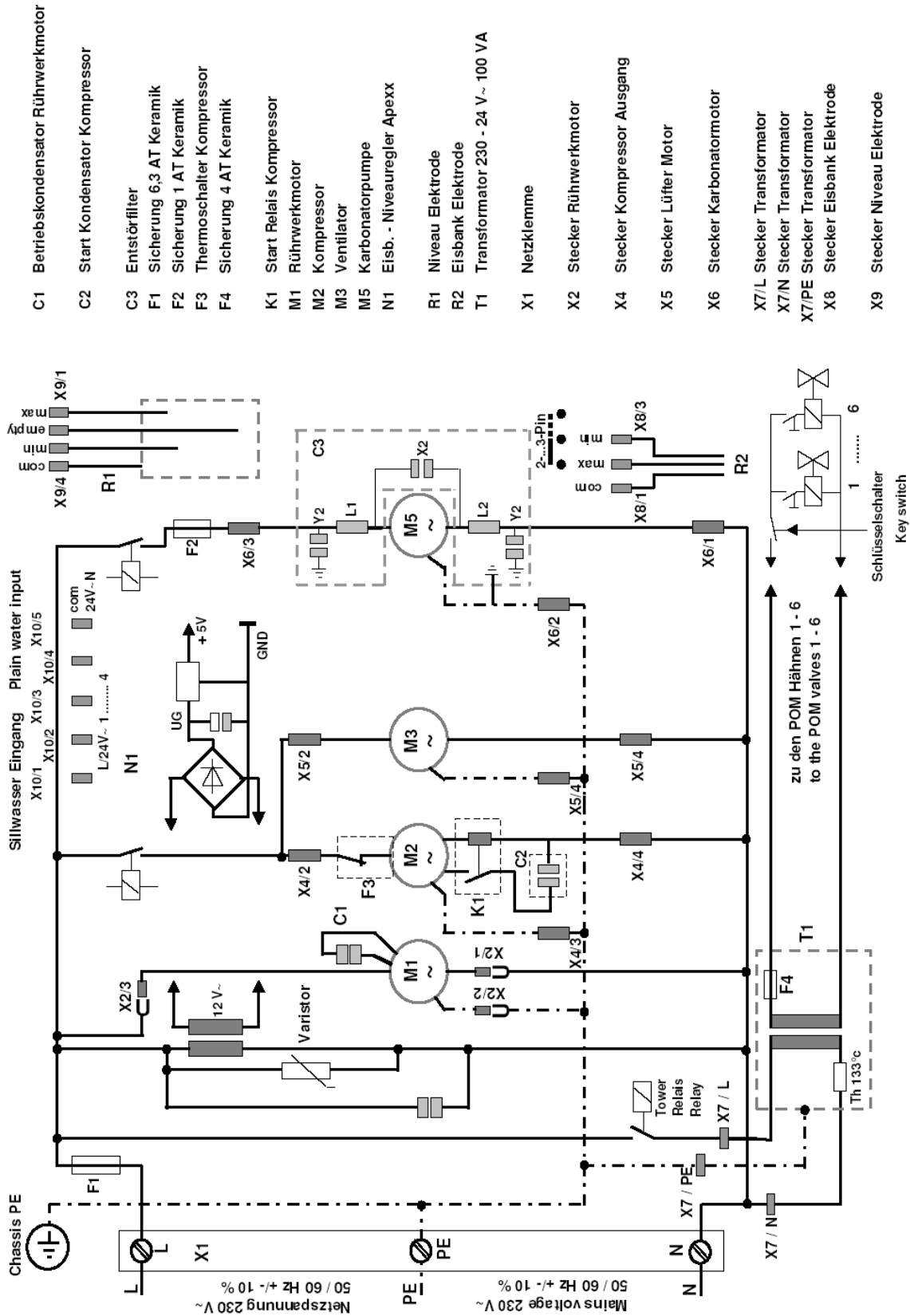
LOOP XL 6 valve Postmix



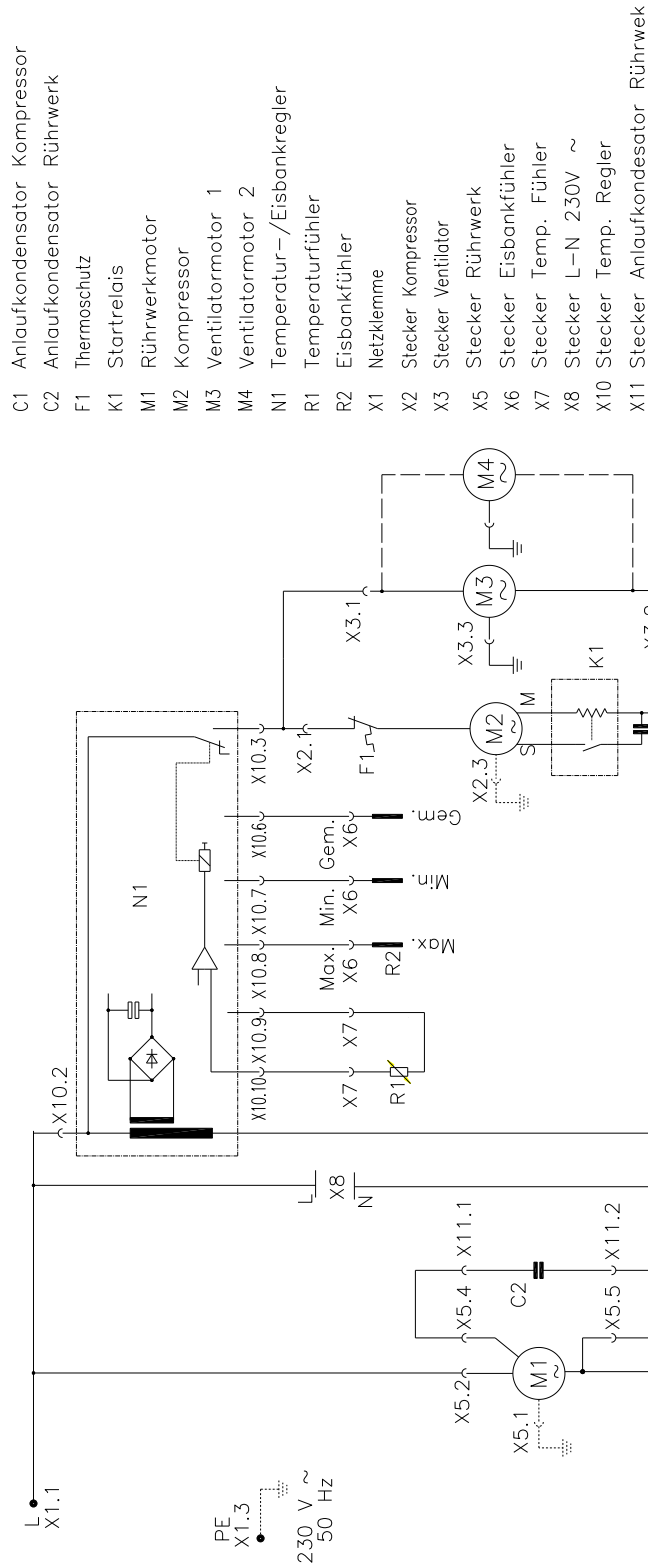
LOOP Premix



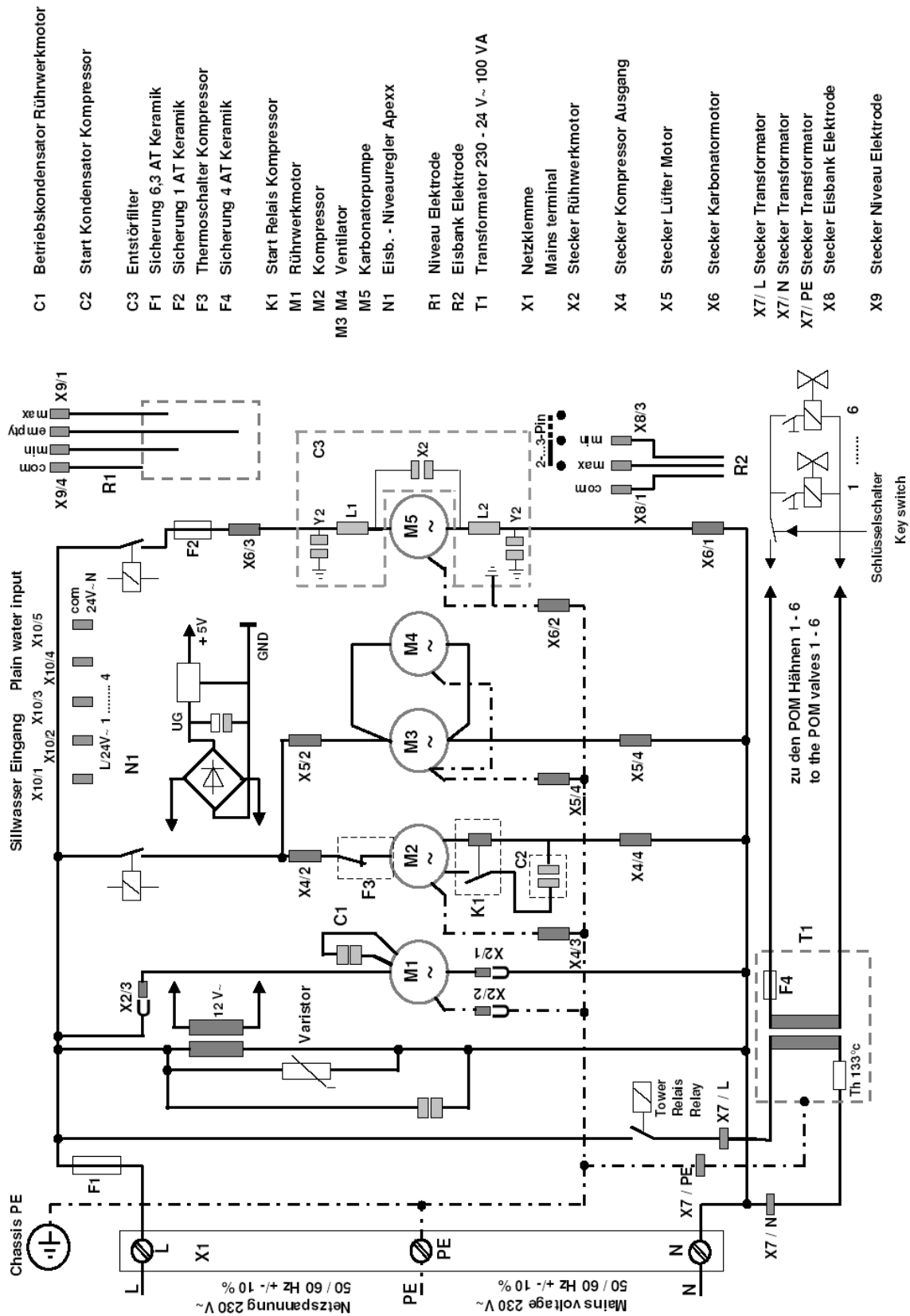
LOOP Postmix



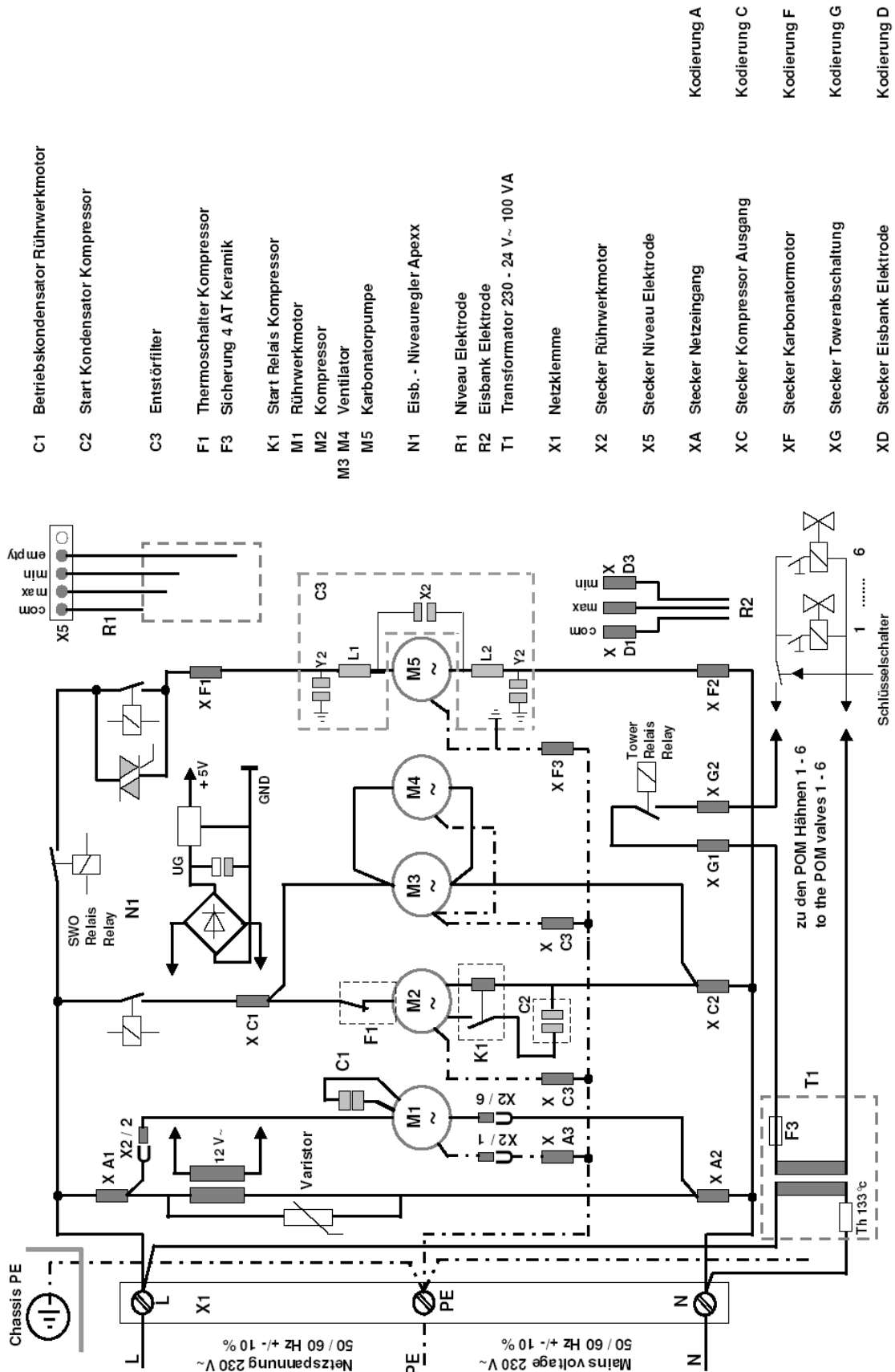
LOOP XL Premix



LOOP XL Postmix



LOOP XL Postmix with Apexx Control



12 Declaration of conformity



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EG - Konformitätserklärung

Im Sinne der EMV - Richtlinie	2004/108/EG
Niederspannungs - Richtlinie	2006/95/EG
Druckgeräte - Richtlinie	97/23/EG
RoHS - Richtlinie	2002/95/EG
Lebensmittelmateriale Richtlinie	1935/2004/EG

Hersteller / Manufacturer:
Gerätebeschreibung / Type of Unit:
Typenbezeichnung / Name of Unit:

Normen, mit denen Konformität bescheinigt wird:

EC - Declaration of Conformity

According to EMC - Directive	2004/108/EG
Low Voltage - Directive	2006/95/EG
Pressure - Directive	97/23/EG
RoHS - Directive	2002/95/EG
Food Contact Material Directive	1935/2004/EC

IMI CORNELIUS DEUTSCHLAND GMBH
Obertheckenkühler / Overcounter Cooling Unit
Loopino, Loop, Loop XL

Regulations under which conformity are certified:

EN 55014 - 1
EN 55014 - 2
EN 61000 - 3 - 2
EN 61000 - 3 - 3
EN 60335 - 1
EN 60335 - 2 - 24
EN 60204 - 1
DIN EN ISO 12100 - 2
DIN EN ISO 14121 - 1

Erklärung :

Hiermit erklären wir, dass die oben genannten Produkte bei bestimmungsgemäßer Verwendung auf Grund ihrer Bauart sowie in den von uns in Verkehr gebrachten Ausführungen den einschlägigen grundlegenden Anforderungen der EG - Richtlinien entsprechen.

Des weiteren erklären wir, dass die in Serie gebauten Geräte dem geprüften Baumuster entsprechen.

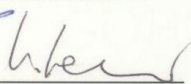
Langenfeld, den / Dated, Langenfeld 28.06.2011



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ppa K. Wiemer
Entwicklungsleiter
Engineering Manager

Amtsgericht Düsseldorf HRB 45002
WEEE-Reg.-Nr. DE26128839
Geschäftsführer:
Andrew Hume
Mark Watts

Declaration :

We certify herewith that all above products comply with the relevant basic requirements of the EC regulations, provided that the products are used in according with their design and purpose as marked by our company.

Furthermore, we certify the units manufactured in series comply with the approved prototype